

EthnoVerse: A Digital Framework for Cultural Archiving

Kaavish Report
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EthnoVerse: A Digital Framework for Cultural Archiving

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Dedication

We dedicate this work to the communities whose stories, traditions, and lived experiences inspired the foundation of this project. Their resilience, cultural richness, and commitment to preserving identity across generations stand at the heart of *EthnoVerse*.

This project is also dedicated to our families, whose constant support, patience, and encouragement made this journey possible.

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Finally, we acknowledge the collaborative efforts within our team. Through continuous discussion, problem-solving, and adaptability, we were able to navigate both technical and conceptual challenges and bring this project to completion.

Abstract

In an age where technology enables the preservation and spread of knowledge, many cultural practices, especially those of marginalized communities, remain undocumented or inaccessible. Our idea for this project grew out of a Tehqiq 2025 ethnographic study on tattoo traditions among Sindh’s Indigenous Hindu Tribes, where we found that living practices were largely absent from public and scholarly archives. This absence reflects a broader gap: there is no centralized, accessible system to document Pakistan’s Indigenous communities as part of its living cultural landscape.

EthnoVerse addresses this gap by creating a web-based repository for multimedia documentation: texts, images, oral histories, and videos organized through searchable metadata.

Beyond static records, the project introduces an experimental 3D module that reconstructs community environments using 3D Gaussian Splatting (3DGS), a recent real-time neural rendering technique that enables efficient and photorealistic scene visualization. This allows users to experience immersive virtual tours with significantly lower latency.

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1. Introduction

1.1 Problem Statement

Indigenous tribal communities in Pakistan are routinely under-represented in public archives and scholarly resources. While valuable, existing archival bodies in Pakistan primarily hold government, colonial, or language-specific records and do not provide accessible, multimedia documentation of contemporary tribal community life. As a result, cultural practices, oral histories, and built environments of these communities remain fragmented, inaccessible, or framed only as historical relics rather than as part of a continuing cultural present. This project addresses the absence of a centralized, searchable, multimedia archive that documents tribal communities in ways that are accessible to researchers, educators, and the public. Our motivation is both technical (to build a robust, extensible system and 3D reconstruction techniques) and social (to make community knowledge discoverable and ethically represented).

1.2 Proposed Solution

We propose a **web-based, extensible Digital Archival System (DAS)** to create a centralized platform to document and make accessible the cultural practices, oral histories, and artistic traditions of tribal communities in Pakistan. The system will allow authenticated administrators to upload multimedia content—text, images, audio, and video—along with metadata for categorization. Public users can browse and search this content through an intuitive, user-friendly interface. The platform aims to bridge the information gap by providing an open, structured, and searchable repository for cultural documentation that is typically fragmented across private archives. For proof of concept, the platform will include a dataset from ethnographic work with the Kolhi community in Tharparkar.

1.3 Intended Users

EthnoVerse will serve multiple types of users, each interacting with the system in different ways:

- **General Public/Researchers, Students, and Scholars:** Individuals seeking to learn about Pakistan’s tribal communities in an accessible and engaging manner. They can browse collections, view images and videos, listen to oral histories, and experience

immersive 3D tours that bring cultural practices to life. Moreover, academics, educators, and students interested in ethnography, anthropology, history, and cultural studies. They can search through tagged data, access detailed metadata for archival material, and utilize the system’s structured categorization for academic research or coursework.

- **Administrators:** Authorized personnel responsible for managing the archive’s content and system operations. They handle user authentication, upload new media, edit or remove existing entries, and ensure the ethical and technical maintenance of the archive. Administrators are also responsible for ensuring data integrity, backups, and system updates.

1.3.1 User Characteristics

- **General Public:** Basic computer literacy; likely to access the system via desktop or mobile browsers. Motivated by curiosity or cultural interest.
- **Researchers/Students:** Moderate to advanced digital literacy; familiar with academic research tools and metadata use. Will require precision search and possibly citation functionality.
- **Administrators:** Advanced technical skills; capable of using web-based admin panels, managing databases, and ensuring data consistency.

1.4 Project Gantt chart and deliverables

Kaavish I Deliverables

- Software Requirement Specification (SRS)
- Software Design Specification (SDS)
- Functional Web Platform Prototype
- Preliminary Dataset Integration (Kolhi community)
- Testing and Evaluation Report

Kaavish II Deliverables

- 3D Reconstruction Module (NeRF / Gaussian Splatting)
- Integrated Virtual Tour Interface
- Optimization and Final Integration of all modules
- Final Report, Presentation, and Deployed System

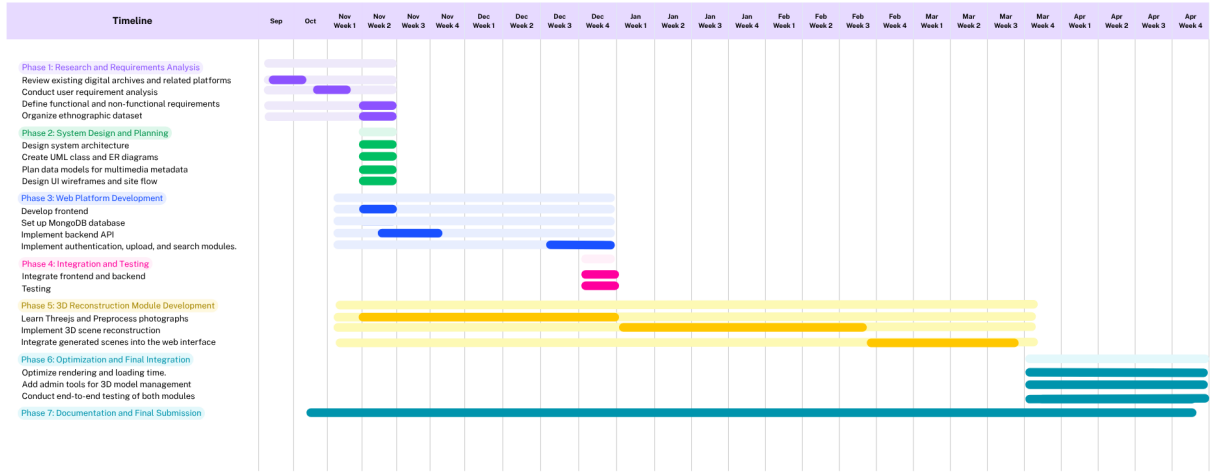


Figure 1.1: Project Gantt Chart for EthnoVerse

1.5 Key Challenges and Mitigations

1.5.1 Technical complexity of 3D reconstruction

- **Risk:** Implementing Gaussian Splatting from scratch is technically demanding and GPU-intensive.
- **Mitigation:** Treat the neural reconstruction as an experimental PoC with a documented fallback (photo-based panoramic/stitched virtual tour). Use preexisting open-source implementations where licensing allows; limit model scope to a single site for the final deliverable if resources/time are constrained.

1.5.2 Data quality & quantity

- **Risk:** Insufficient or uneven photographic coverage can prevent accurate 3D reconstruction or reduce media utility.
- **Mitigation:** Specify photo capture guidelines, and request additional images from field collaborators.

1.5.3 Compute resources

- **Risk:** Training or running reconstruction models may require more GPU capacity than available on local machines.
- **Mitigation:** Use lightweight model variants, perform experimental runs on cloud credits, or restrict training to smaller scene subsets. Document exact hardware used so results are reproducible.

1.5.4 Scope creep & time management

- **Risk:** Ambitious goals (multi-community coverage, advanced search, full 3D) could exceed Kaavish timelines.
- **Mitigation:** Use the Kolhi dataset as a bounded PoC; prioritise core archival functionality (ingestion, metadata, search, admin) during Kaavish I and reserve advanced features for Kaavish II. Maintain weekly sprints and milestone checks.

2. Literature Review

Digital archiving has become an important research area at the intersection of Computer Science, Digital Humanities, Human Computer Interaction (HCI) and Cultural & Historical Studies. With increased recognition of the need to document marginalized and Indigenous cultures, scholars have developed digital platforms and computational tools to preserve oral histories, visual records, and living practices. This chapter reviews existing work on local digital archives (in the context of Pakistan), multimedia repositories, and 3D reconstruction methods, and identifies the limitations that motivate the development of *EthnoVerse*. It also highlights the novelty of the platform in comparison with state-of-the-art systems.

2.1 Digital Archives

Digital archives represent a paradigmatic shift in the preservation and dissemination of cultural memory. Prior to digitization, archival continuity relied on oral transmission, manuscript circulation, and material reproduction across generations. Studies note that archival practice—understood broadly as the cultural act of transmitting memory—has long preceded institutionalized archives and remains deeply embedded in human social organization [18, 6]. The emergence of digital repositories has expanded archival functionality through networked databases, distributed storage, and cloud infrastructures, enabling long-term preservation, metadata-driven access, and multimodal documentation.

Scholars argue that digital archives democratize access but simultaneously reproduce questions of authority, authenticity, and representation, particularly when dealing with marginalized or non-textual knowledge systems [10, 8]. Public digital libraries, including DigiLibraries.com [5] and the Higher Education Commission Digital Library of Pakistan [9], demonstrate the global proliferation of digital archival infrastructures, yet these platforms largely reproduce textual epistemologies. Contemporary archival scholarship emphasizes the need for multimodal, community-centered, and ethically governed digital environments capable of accommodating audiovisual, spatial, and embodied cultural expressions.

2.1.1 Community Archiving on the Global Scale

Community archiving has gained substantial scholarly attention as a counter-archival practice that centers the self-representation of communities historically marginalized by institutional memory regimes. Flinn defines community archives as grassroots initiatives in which communities “take power into their own hands” to preserve their histories, identities, and

cultural heritage independent of established archival institutions [7]. Gilliland and Flinn further emphasize that community archives challenge dominant archival narratives by producing alternative forms of evidence, memory, and identity [13].

Internationally, numerous initiatives document community life, diasporic histories, and local memory. The Community Archives and Heritage Group (CAHG) [3] consolidates archives produced by diverse organizations in the UK and Ireland, including the Archive of the Association of Ukrainian Women in Great Britain, the Living Refugee Archive [11], and Windrush community archives. Social media platforms have also emerged as vernacular archives, particularly among youth cultures. Rosales’s projects *Veteranas and Rucas* and *Map Pointz* document visual histories of Latinx youth subcultures in Los Angeles, addressing erasure through participatory digital archiving [17]. Other significant community archives include *A People’s Archive of Police Violence in Cleveland* [1], *Densho*—an archive documenting Japanese American incarceration during World War II [4]—and *Loss/Capture*, a contemporary visual narrative archive [12].

These projects illustrate how digital tools enable communities to construct their own evidentiary and historical record, challenging institutional biases and expanding archival epistemologies toward participatory, polyvocal, and socially accountable memory work.

2.1.2 Existing Local Archival Work

Pakistan’s archival landscape remains limited in scope, accessibility, and inclusivity. While several institutions preserve literary works, governmental documents, and cultural heritage, these archives prioritize textual, bureaucratic, or elite historical narratives. Studies highlight that marginalized and indigenous communities remain significantly underrepresented within national heritage infrastructures. The lack of multimodal, community-centered digital archives in Pakistan contributes to the persistent misrepresentation of indigenous groups as peripheral or historically displaced.

Notable Pakistani archival initiatives include the 1947 Partition Archive [20], a major oral history repository documenting Partition survivors, and the Citizens Archive of Pakistan (CAP) [2], a nonprofit institution preserving photographic, oral, and material histories of Pakistan’s postcolonial development. However, both initiatives primarily document macro-historical narratives or urban middle-class experiences. CAP also primarily houses colonial manuscripts, administrative records, and legal documents, with minimal representation of tribal or nomadic communities. The E-Library of Punjab [15] similarly centralizes academic and literary materials rather than ethnographic or community-generated sources.

2.1.3 Limitations, Gaps & Competitor Analysis of Existing Archives

The structural absence mentioned above produces two interrelated gaps:

- indigenous communities are excluded from dominant narratives of Pakistani cultural identity, and
- researchers, policymakers, and the public lack accessible platforms for engaging with these communities’ cultural practices, socio-historical contexts, and everyday lived realities.

Moreover, most archival institutions in Pakistan primarily preserve textual or bureaucratic documents. These archives often lack:

- public searchability and modern user interfaces,
- community-centered metadata practices,
- immersive or spatial representations of cultural environments.

The 1947 Partition Archive

The 1947 Partition Archive is one of the largest oral history collections in South Asia, documenting the experiences of Partition survivors through crowdsourced testimonies [20]. Its contributions to public history are significant; however, its thematic scope is narrowly restricted to the events, memories, and aftermath of Partition. As a result, it excludes the historical trajectories, cultural practices, and contemporary lifeworlds of indigenous tribal groups.

Gap: *The archive focuses exclusively on Partition-era migration narratives and does not include ethnographic, cultural, or spatial documentation of indigenous communities in Pakistan.*

Citizens Archive of Pakistan (CAP)

This is the largest governmental archival body in the province, housing colonial records, judicial correspondence, land documents, and administrative manuscripts. Its collections remain largely inaccessible to the public and are oriented toward bureaucratic recordkeeping rather than cultural documentation.

Gap: *The repository contains virtually no material on indigenous tribes in Sindh, thereby structurally excluding marginalized communities from the state's archival memory.*

Punjab Digital Library (E-Library Punjab)

The Punjab Digital Library focuses primarily on digitized manuscripts, academic texts, and historical documents [15]. While expansive in scope, the platform is text-centric and does not support ethnographic or multimedia forms of documentation.

Gap: *Its emphasis on textual heritage excludes oral histories, cultural practices, community-generated knowledge, or spatial representations of indigenous groups.*

South Asian American Digital Archive (SAADA)

SAADA is a prominent digital archive centered on the South Asian diaspora in North America [19]. Although methodologically relevant, it does not address communities situated within Pakistan.

Gap: *SAADA's geographic and thematic scope precludes documentation of indigenous tribal populations within Pakistan, making it unsuitable as a local archival resource.*

This absence of a centralized, multimedia, accessible system for Indigenous communities highlights a critical infrastructural gap that *EthnoVerse* aims to address.

Archive Name		Gap Description	Type	
1947 Partition Archive		Focuses exclusively on Partition-era migration narratives; lacks ethnographic, cultural, or spatial documentation of indigenous communities within Pakistan.	Oral History	
Citizens of Pakistan (CAP)	Archive	Contains colonial/state administrative records; virtually no material on indigenous tribes. Restricted public access prevents community engagement and visibility.	Government Records	
Punjab Library	Digital	Text- and manuscript-centered repository; excludes oral traditions, cultural practices, audiovisual material, and community-generated knowledge.	Manuscripts, Books	
SAADA		Focused on the South Asian diaspora in North America; geographically and thematically irrelevant for documenting indigenous communities in Pakistan.	Diaspora Archive	

Table 2.1: Comparison of Local Archives.

2.2 3D Reconstruction for Cultural Heritage

2.2.1 Photogrammetry and Early 3D Techniques

Photogrammetry has historically been used to digitally reconstruct archaeological or architectural sites [16]. However, these techniques rely heavily on controlled lighting, dense image capture, and consistent camera conditions—constraints that are difficult to maintain.

2.2.2 Neural Radiance Fields (NeRF)

Neural Radiance Fields (NeRF), introduced by Mildenhall et al. [14], represent one of the most significant recent advancements in neural rendering and view synthesis. NeRF reconstructs a continuous 3D scene representation by optimizing a neural network to map spatial coordinates and viewing directions to volumetric density and emitted color. This enables photorealistic novel-view synthesis from sparse image sets—an essential capability for cultural heritage documentation, where controlled scanning conditions are rarely available. In-depth working of the algorithm is discussed in the SDS chapter.

Advantages for Cultural and Ethnographic Reconstruction

NeRF offers several critical advantages for documenting indigenous and cultural environments:

- **High fidelity from sparse inputs:** Accurate reconstructions even with limited, imperfect, or unstructured image collections.
- **Viewpoint continuity:** Produces smooth novel viewpoints, enabling immersive 3D virtual walkthroughs.
- **Realistic lighting modeling:** Positional encodings allow NeRF to reproduce complex textures and structures found in cultural artifacts.
- **Non-intrusive capture:** Requires only RGB images—ideal for sensitive cultural settings where scanning hardware (e.g., LiDAR) is impractical.

2.2.3 Gaussian Splatting (3DGS)

3D Gaussian Splatting (3DGS) represents a recent advancement in real-time radiance field rendering, introduced as an alternative to computationally intensive NeRF models. Instead of encoding the scene in a neural network, 3DGS represents it as a collection of anisotropic Gaussian primitives in 3D space. Each Gaussian encodes position, color, opacity, and covariance, allowing efficient rasterization directly on the GPU. Compared to NeRF, 3DGS offers:

- **Real-time rendering:** Enables interactive frame rates suitable for web-based visualization
- **Faster training times:** Scene optimization completes in minutes instead of hours
- **Explicit representation:** Easier integration with graphics pipelines like Three.js
- **Lower computational overhead:** More feasible for mid-range GPU setups

2.3 Novelty and Contribution of *EthnoVerse*

EthnoVerse contributes to the state-of-the-art by:

- creating Pakistan’s first multimedia, searchable digital archive dedicated to Indigenous communities,
- integrating state-of-the-art Gaussian Splatting methods within a cultural archival workflow,
- designing a scalable platform capable of supporting multiple communities,

- embedding ethical, consent-driven metadata aligned with Indigenous sovereignty frameworks,
- merging ethnographic research methodologies with full-stack engineering and computer vision techniques.

Unlike prior archives, *EthnoVerse* emphasizes the representation of *contemporary* community life and provides immersive virtual tours of cultural spaces rather than static documentation.

3. Software Requirement Specification (SRS)

This Software Requirements Specification (SRS) outlines the software and system requirements for EthnoVerse. It details functional and non-functional requirements for system capabilities. The document also includes external interfaces to clarify the system’s design and interactions.

3.1 Functional Requirements

This section specifies the functional requirements of the system. The requirements are divided into two primary categories: those pertaining to the **web platform** and those related to the **3D virtual tour module** developed using Gaussian Splatting. Each subsection includes a tabular summary and a module-wise functional outline for clarity.

3.1.1 EthnoVerse Web Platform

The web platform is responsible for storing, displaying, and managing multimedia data related to community archives. Only administrators have access to content management features, while public users can freely view and search content without authentication.

- **Module 1: Admin Authentication**
 - Secure login and session management.
 - Only authenticated admins can perform CRUD operations on content.
- **Module 2: Archive Content Management**
 - Upload of multimedia items with validated metadata.
 - Edit or delete entries as required, with logs maintained for accountability.
 - Metadata management supporting version control and validation.
- **Module 3: Search and Discovery**
 - Keyword search across titles, tags, and descriptions.
 - Filtering by community, content type, or upload date.

- **Module 4: Content Presentation**
 - Responsive user interface (React.js) displaying media grid and detailed views.
 - Embedded media players for image, audio, and video files.
- **Module 5: System Administration**
 - Dashboard summarizing total uploads and user activity.
 - Automated data backups and integrity verification.
 - Logs for uploads, edits, deletions, and admin sessions.
- **Module 6: System Extensibility and Scalability**
 - Supports addition of new communities and datasets without structural changes.
 - Allows integration of new media types through modular schema design.
 - Ensures smooth performance as data volume and content grow.

ID	Function Name	Description	Users
FR-W01	Admin Authentication	Admins log in securely using authentication to access upload, edit, and delete functionalities.	Admin
FR-W02	Media Upload	Admin uploads multimedia content (text, image, audio, video) with metadata (title, tags, community, description, consent).	Admin
FR-W03	Edit/Delete Content	Admin edits metadata or removes outdated/incorrect items. All edits are logged.	Admin
FR-W04	Metadata Management	Validates and stores metadata; supports updates and versioning.	Admin
FR-W05	Search Content	Users search items by title, tags, or description.	Admin, Public
FR-W06	Filter Results	Users filter items by community, media type, or upload date.	Admin, Public
FR-W07	View Media	Public users view multimedia content through responsive layouts.	Public
FR-W08	Dashboard Overview	Admin dashboard displays statistics such as total uploads and media distribution.	Admin
FR-W09	Backup and Data Integrity	Automated database backups and consistency checks ensure data reliability.	System
FR-W10	Logs and Audit Trail	Records admin actions (login, upload, edit, delete) for transparency.	System, Admin
FR-W11	Scalability and Generalization	System supports addition of new communities without structural changes or degradation. Database schema and API design ensure scalability and modular extensibility.	System, Admin

Table 3.1: Summary of Functional requirements for EthnoVerse web platform.

3.1.2 3D Reconstruction and Virtual Tour

The 3D module demonstrates the technical application of computer graphics techniques to reconstruct community environments from photographs.

- **Module 1: Data Preparation**

- Capture or collect photographs of community sites (courtyards, huts, shrines) from multiple angles.
- Preprocess images (resize, align, normalize lighting) before 3DGS input.

- **Module 2: Scene Reconstruction**

- Implement 3D Gaussian Splatting (primary) with NeRF explored as a baseline method.
- Generate scene models stored in compressed formats for web use.

- **Module 3: Rendering and Visualization**

- Convert reconstructed 3D scenes into Three.js environments.
- Embed virtual tours within the frontend interface.

- **Module 4: Scene Management and Maintenance**

- Store reconstructed models and associated metadata (scene name, date, contributors).
- Allow admin to update or delete 3D scenes.
- Implement caching and loading optimization for performance.

- **Module 5: Fallback and Error Handling**

- If reconstruction fails or resources are insufficient, generate a static 360° panoramic viewer.
- Log all reconstruction processes and errors for debugging.

ID	Function Name	Description	Users
FR-3D01	Dataset Preparation	Collect and preprocess multiple-angle photographs of community spaces for Gaussian Splats input.	Admin
FR-3D02	Scene Reconstruction	Generate 3D radiance field representations using Gaussian Splatting techniques.	System
FR-3D03	Rendering and Visualization	Render 3D scenes into navigable virtual tours integrated into the web interface (via Three.js).	Public, Admin
FR-3D04	Scene Management	Store, update, or remove 3D models and their metadata (scene name, location, source images).	Admin
FR-3D05	Performance Optimization	Optimize rendering quality and load times for web deployment.	System
FR-3D06	Fallback Handling	If reconstruction fails, fallback to photo-based 360° panoramic tour is generated.	System

Table 3.2: Summary of Functional requirements for the 3D reconstruction module.

3.2 Non-Functional Requirements

The following non-functional requirements ensure the performance, reliability, scalability, and ethical compliance of the EthnoVerse system.

- **Performance:**

- The system must support at least 50 concurrent users with average response times under 2 seconds for typical queries.
- Search results should render within 1–3 seconds for datasets containing up to 10,000 items.
- Media files must be optimized for web display using compressed images and adaptive video streaming techniques.

- **Reliability:**

- The database must maintain at least 99% uptime under normal conditions.
- Failed upload sessions should automatically retry or roll back transactions to preserve data integrity.
- The system should recover gracefully from unexpected shutdowns or network interruptions.

- **Data Integrity and Backup:**

- Automatic weekly database backups will be maintained for both local and cloud storage.
- Metadata will use version control to track all edits and revisions.
- Redundant storage solutions will ensure that no data is lost in case of system failure.

- **Usability:**

- The user interface must follow WCAG 2.1 accessibility standards and maintain a responsive design across devices.
- The upload interface should include clearly labeled fields and drag-and-drop functionality.
- Search and filter options must remain visible and accessible on all main views.

- **Compliance and Ethics:**

- Ethical representation of indigenous content will be ensured with support from an SDP professor and student volunteer specializing in ethnographic research.
- Administrators must verify consent documentation before publishing any community media.
- Cultural materials will be presented with appropriate attribution and context.

3.3 External Interfaces

3.3.1 User Interfaces

This section includes our mockup screens and briefly explains them. The platform provides two main user-facing interfaces:

- **Public Interface (Web Portal):**
 - Accessible via web browser.
 - Features a home page with search bar, category filters, and featured archives.
 - Each media item has a detail page with description, metadata, and related items.
 - 3D tour viewer embedded within the page for immersive experiences.
 - **Admin Interface:**
 - Accessible only to authorized users.
 - Includes upload form, metadata editor, user management, and backup controls.
-



Figure 3.1: Homepage: Living Archives Main Landing Screen.

This is the initial entry point, providing an immediate overview of the project's mission as a digital storytelling platform and cultural archive. The page presents the system's core features: a brief mission statement, a featured community (Kolhi), and primary navigation buttons (**Explore Communities** and **Search Archives**). Users are led directly to the browse or search functionalities.

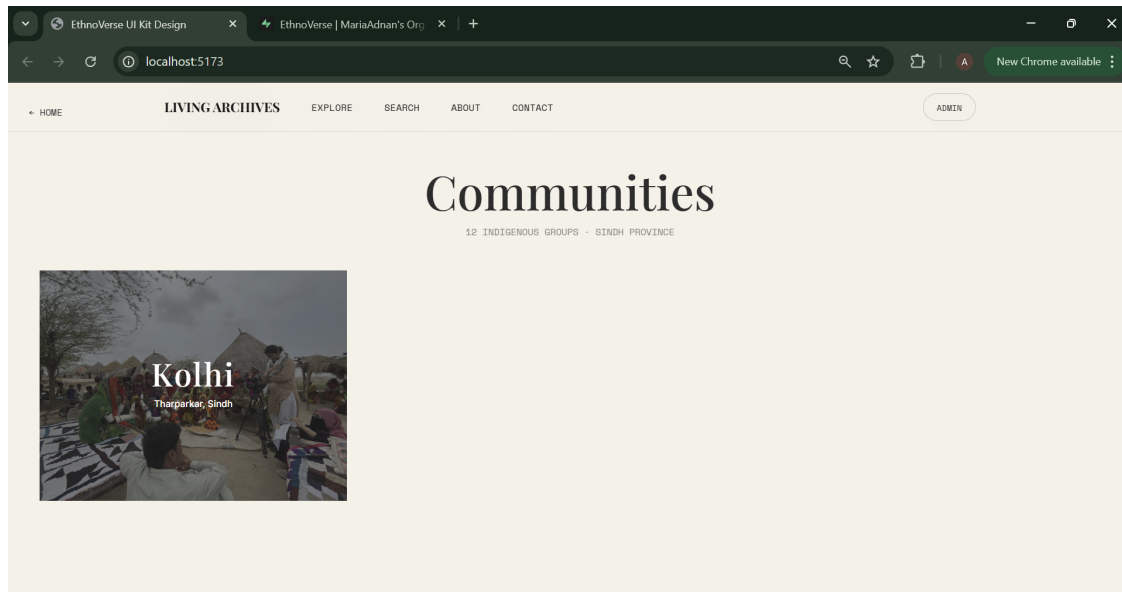


Figure 3.2: Explore Communities View.

This view allows users (Public/Researchers) to browse the archive based on documented community groups. The interface displays a card-based list of all communities currently archived (e.g., Kolhi, Example Community). Each card includes the community name, location (Tharparkar), a brief introduction, and a link to **View Details**, which directs the user to the dedicated Community Detail Page.

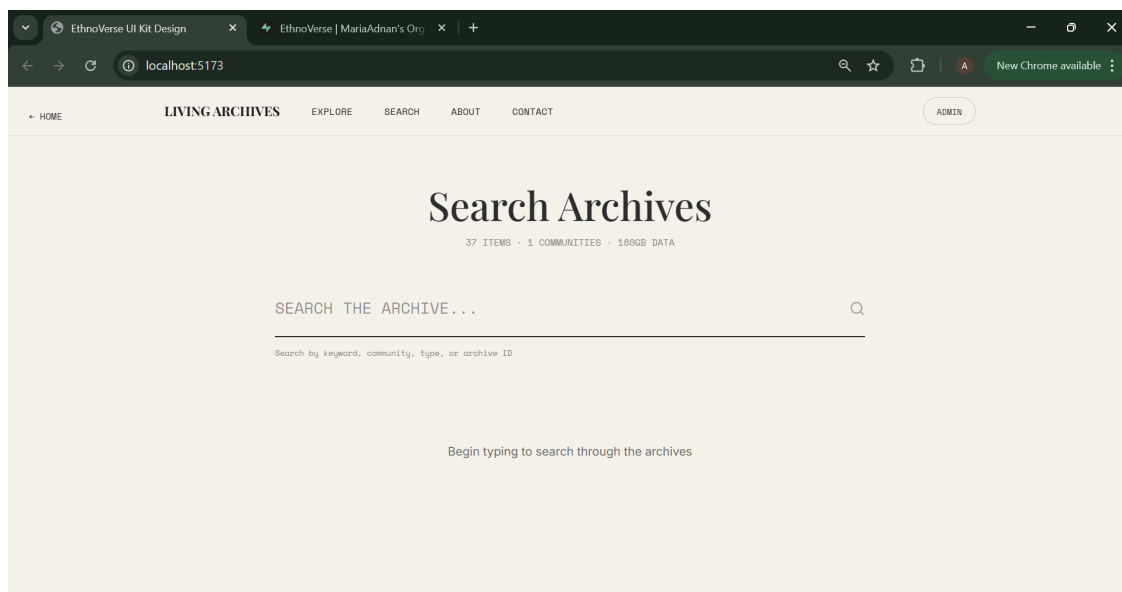


Figure 3.3: Search Communities View.

The Search View enables users to efficiently find specific communities or media based on keywords, tags, or descriptive metadata. This page features a universal search bar across all

content. The results are presented in cards, and the underlying logic applies to individual media titles and tags to support deep discovery. Users can refine their search by employing filters based on community, content type, or upload date

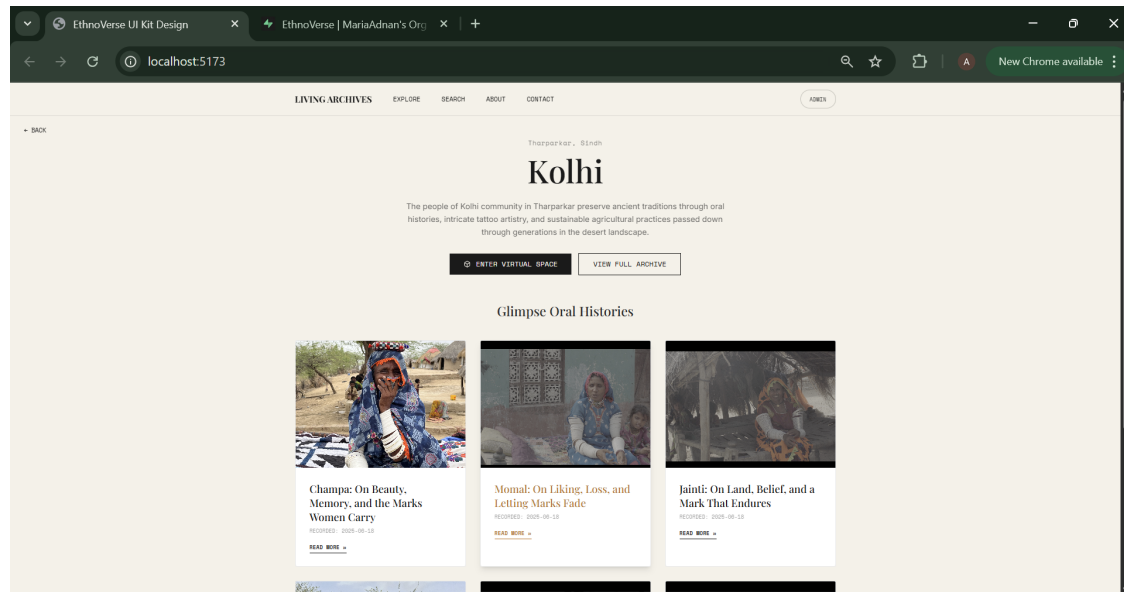


Figure 3.4: Community Detail Page (Kolhi).

This page serves as the dedicated hub for a specific community, showcasing all available content (multimedia and 3D) for spatial immersion. The header provides context and two primary action buttons: **Open 3D Tour** (linking to the immersive view) and **View Media** (linking to individual content). The lower section, **Media Collection**, displays a grid of multimedia items (Image, Audio, Video). This page essentially bridges the archival content with the immersive virtual tour experience



Figure 3.5: 3D Virtual Tour Viewer.

The viewer provides an immersive, navigable virtual tour experience of the reconstructed community environment. The screen hosts the interactive 3D rendering, generated using Gaussian Splatting, which is optimized for web deployment. A supporting section, **About This Tour**, provides context on the ethnographic and technical basis of the reconstruction.

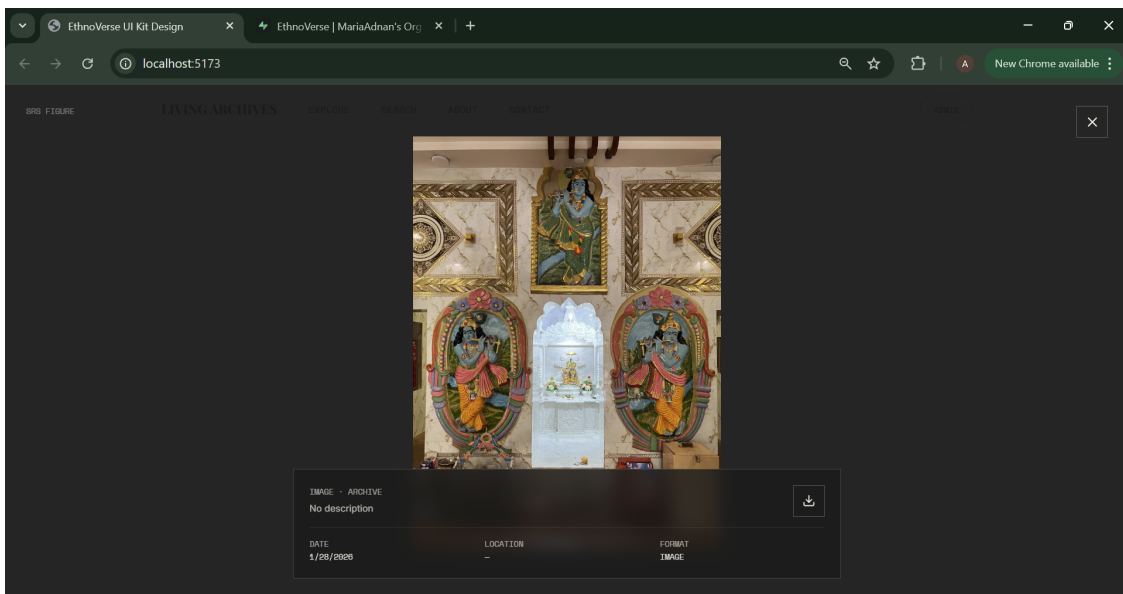


Figure 3.6: Media Detail Page (Image).

This page in particular displays the specific image (e.g., **Traditional Tattoo**) and indicates its media type (**IMAGE**) and collection.

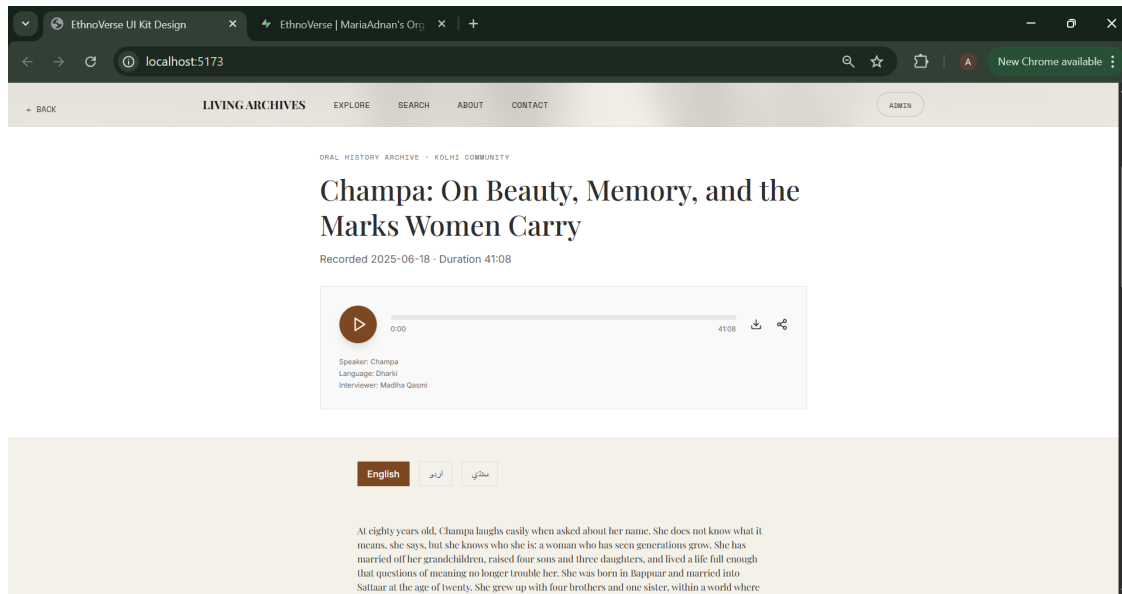


Figure 3.7: Media Detail Page (Audio/Oral History).

The interface presents audio content (oral histories, songs) and makes it searchable and accessible through transcription. The view features an embedded audio player and a dedicated **Transcript** panel below it. The core function is to allow users to search the transcripts for keywords and potentially jump to corresponding timestamps in the media.

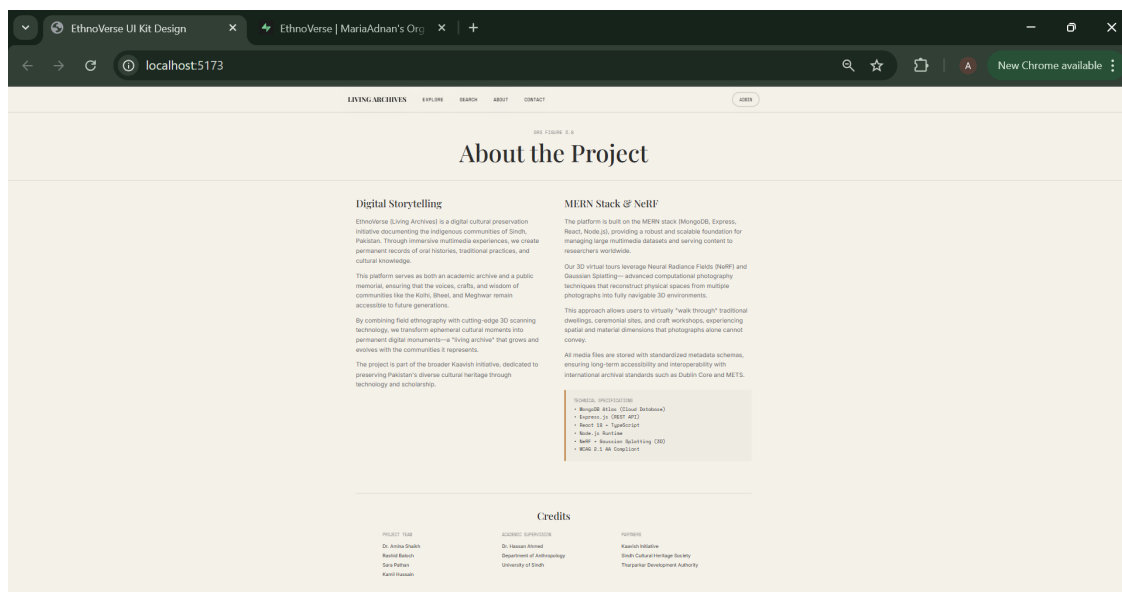


Figure 3.8: About Living Archives Page.

This page provides background and rationale to stakeholders, researchers, and the public. This page details the project's roots as a Kaavish initiative, its technical foundation (MERN

stack integration), and its commitment to preserving the heritage of Sindh’s communities. Its inclusion ensures the user is informed about the project’s scope and technology, aligning with the project’s goal of openness.

3.3.2 Application Program Interface (API)

The system is built using a **RESTful API** architecture (Node.js/Express backend) to manage data flow between the React frontend, the Supabase database, and the Python-based 3D Reconstruction Engine.

Core API Endpoints

- **Auth Endpoints:** Facilitate user login and session management (`/auth/login`, `/auth/register`) using **JWT** for secure, role-based access control
- **Content Management Endpoints:** Restricted to Administrators, enabling CRUD operations:
 - `POST /api/media`: Upload new multimedia content (image, audio, video) with metadata.
 - `PUT /api/media/{id}`, `DELETE /api/media/{id}`: Edit or remove existing archive entries.
- **Search and Discovery Endpoints:** Publicly accessible read operations:
 - `GET /api/communities`: Retrieve a list of all documented communities.
 - `GET /api/media?q=keyword&community=kolhi`: Search content based on keywords and filter by metadata.
- **3D Scene Endpoints:** Used for the virtual tour module:
 - `POST /api/3d/reconstruct`: (Admin/System) Triggers the Gaussian Splatting engine for a new scene reconstruction.
 - `GET /api/3d/{sceneId}/model`: Retrieves the optimized 3D model data (e.g., JSON, point cloud) for rendering in the client’s Three.js viewer.

3.3.3 Hardware and Communication Interfaces

- **Client-Side:** Standard desktop or laptop computers with modern web browser compatibility.
- **Server-Side Infrastructure:** The backend and database are hosted on a cloud environment, supporting modular scaling.
- **Specialized Hardware (3D Reconstruction):** A mid-range GPU (e.g., NVIDIA RTX 3060 or equivalent) is recommended for 3D reconstruction tasks, managed by a dedicated service or cloud instance (Non-Functional Requirement: Performance).
- **Communication Protocols:** All data transfer, including API calls, media streaming, and asset retrieval, uses HTTPS to ensure data integrity and security.

3.4 Use Cases

3.4.1 Use Case Diagram

The Use Case Diagram visually represents the primary capabilities of the EthnoVerse system and the relationship between the system and its key actors: the Public User/Researcher and the Administrator.

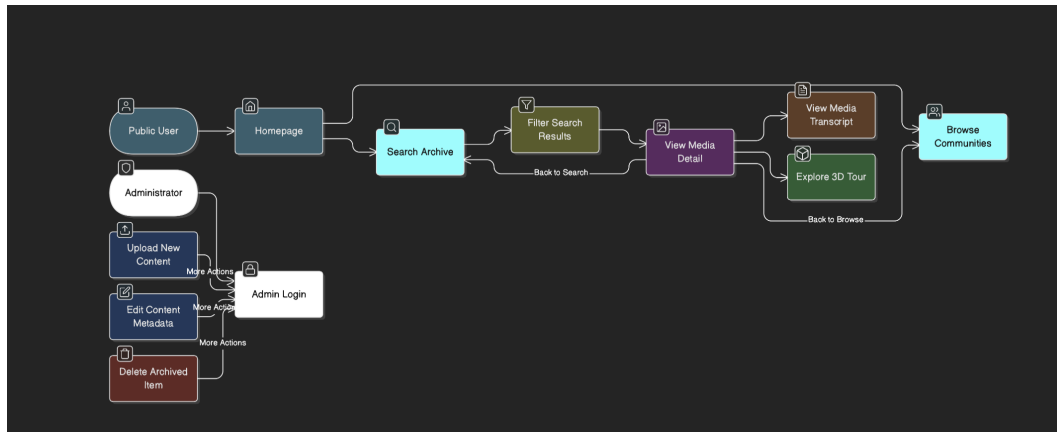


Figure 3.9: EthnoVerse High-Level Use Case Diagram.

Actors:

- **Public User / Researcher:** Individuals seeking to consume content, browse, search, view media, and explore immersive 3D tours.
- **Administrator:** Authorized personnel responsible for content management, technical maintenance, and data integrity.

3.4.2 Use Cases by Actor

Public User / Researcher Use Cases

These use cases detail the primary ways users interact with the public-facing interface, aligning with the core goals of exploration and access.

ID	Use Case Name	Description
UC-P01	Access Homepage	As a user , I want to view the homepage so that I can get an overview of the archive’s mission and easily navigate to its main features like Explore and Search.
UC-P02	Browse Communities	As a user , I want to browse a list of all available communities so that I can discover content by cultural or regional grouping.
UC-P03	Search Archive	As a user , I want to search the entire archive using keywords so that I can find specific media items based on titles, tags, or descriptions.
UC-P04	Filter Search Results	As a user , I want to filter my search results so that I can narrow down the list by community, media type, or date to find exactly what I’m looking for.
UC-P05	View Media Detail	As a user , I want to view a detailed page for a media item so that I can see the content, its full description, metadata, and attribution.
UC-P06	View Media Transcript	As a user , I want to view the transcript for audio or video content so that I can read the text, improve accessibility, and search for specific parts within the media.
UC-P07	Explore 3D Tour	As a user , I want to launch and navigate a 3D virtual tour so that I can immersively explore a reconstructed community environment.

Table 3.3: Use Cases for Public User / Researcher.

Administrator Use Cases

These use cases focus on the control, maintenance, and archival duties performed by authenticated administrators through the secure Admin Interface.

ID	Use Case Name	Description
UC-A01	Secure Login	As an administrator , I want to securely log in to the system so that I can access the protected content management features.
UC-A02	Upload New Content	As an administrator , I want to upload new media files and their metadata so that I can add new cultural documentation to the archive.
UC-A03	Edit Content Metadata	As an administrator , I want to edit the metadata of existing items so that I can correct errors or update information to ensure data accuracy.
UC-A04	Delete Archived Item	As an administrator , I want to delete outdated or non-compliant content so that I can maintain the ethical and factual integrity of the public archive.

Table 3.4: Use Cases for Administrator.

3.5 System Diagram

The EthnoVerse architecture follows a modular, service-oriented design built with a dedicated 3D reconstruction module. Figure 3.10 illustrates the high-level components and data flow between them.

1. Frontend (React.js)

The frontend provides a dynamic, responsive user interface that allows users to browse communities, view multimedia archives, and launch immersive 3D tours. It communicates with the backend through RESTful APIs over HTTPS. Administrators access a separate dashboard for content management, uploads, and system monitoring.

2. Backend (Node.js + Express)

The backend serves as the core logic layer responsible for handling API requests, authentication, routing, and data processing. It manages CRUD operations for multimedia content and interacts with both the database and the 3D reconstruction module. It implements secure JWT-based authentication and role-based access control for administrators.

3. Database (Supabase)

Supabase stores all structured metadata, user profiles, and references to media files. The schema supports flexible, community-specific expansion and version-controlled updates. It also logs administrative actions and maintains backup copies for recovery and integrity checks.

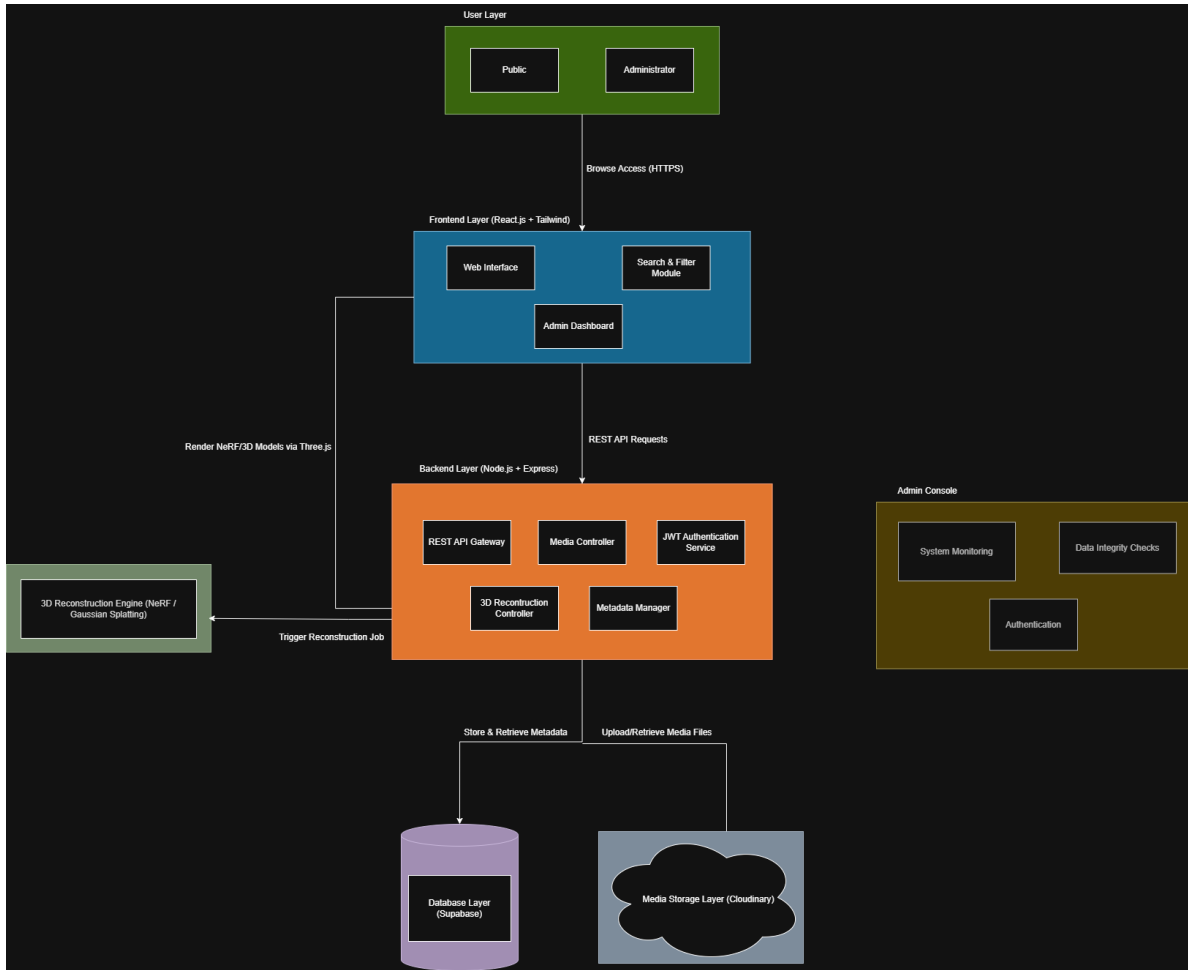


Figure 3.10: EthnoVerse System Block Diagram

4. Media Storage Service

All uploaded media—images, audio, video, and 3D assets—are stored in a cloud-based repository. Metadata in Supabase links each file to its respective community and media type. The storage layer supports redundancy and retrieval optimization for faster access and rendering.

5. 3D Reconstruction Engine (Gaussian Splatting)

This module operates as an independent service responsible for generating immersive 3D scenes from multi-angle photographs. Once processed, the resulting models are optimized for web use and integrated into the platform’s frontend via Three.js. The engine can fall back to 360° panoramic tours if GPU or data constraints arise.

6. Admin Console

The Admin Console provides authenticated users with control over system content and maintenance. It enables secure uploads, metadata edits, deletion of outdated items, and perfor-

mance monitoring. This module also schedules automated database backups and generates logs for transparency.

7. Data Flow Summary

- Users interact with the system through the React.js interface.
- The frontend communicates with the Node.js backend via RESTful API calls.
- The backend retrieves or updates records in Supabase and media storage.
- For 3D content, the backend triggers the reconstruction engine and retrieves the processed model.
- The processed 3D scenes and media are delivered back to the frontend for visualization.

4. Software Design Specification (SDS)

This chapter documents the system design for *EthnoVerse*, describing the software architecture, UML-class-level design, and technical algorithms.

4.1 Software Design



Figure 4.1: UML Class Diagram

4.1.1 Class Description

Class: User

Attributes:

- **userId**: Unique identifier for the user.
- **username**: Display name chosen by the user.
- **email**: Contact email used for login and notifications.
- **hashedPassword**: Encrypted password stored securely.

- **role**: Access level of the user (`admin`, `editor`, `public`).
- **createdAt**: Timestamp indicating when the account was created.
- **profileMeta**: Optional metadata such as biography or affiliation.

Methods:

- **authenticate(email: string, password: string): boolean** – Validates user credentials.
- **authorize(action: string): boolean** – Checks if the user has permissions for a given system action.
- **updateProfile(profileMeta: dict): void** – Updates the user’s metadata.
- **logAction(actionType: string, targetId: string): void** – Records an action in the audit log.

Class: Community

Attributes:

- **communityId**: Unique identifier for the community.
- **name**: Name of the community.
- **location**: Geographic or regional information.
- **shortDescription**: Brief summary of the community.
- **longDescription**: Detailed cultural, historical, or sociological description.
- **language**: Primary language(s) spoken by the community.
- **contactInfo**: Contact details for community representatives or researchers.
- **createdAt**: Timestamp when the community record was created.

Methods:

- **addMedia(mediaId: string): void** – Associates a new media item with the community.
- **updateInfo(details: dict): void** – Updates descriptive and organizational information.
- **getSummary(): dict** – Returns a short, structured summary of the community.

Class: MediaItem

Attributes:

- **mediaId**: Unique identifier for the media file.
- **communityId**: Foreign key linking to the related community.
- **uploaderId**: Identifier of the user who uploaded the item.
- **mediaType**: Type of media (**image**, **audio**, **video**, **document**).
- **storageUri**: File path or cloud URI where the media is stored.
- **thumbnailUri**: Auto-generated thumbnail link.
- **title**: Title of the media item.
- **description**: Detailed contextual description.
- **tags**: List of descriptive tags for indexing and search.
- **dateCaptured**: Date when the media was originally recorded.
- **consentDocUri**: URI to consent or permissions documentation.
- **visible**: Boolean indicating whether the media is publicly visible.
- **createdAt**: Upload timestamp.

Methods:

- **validateMetadata(): boolean** – Ensures metadata completeness and correctness.
- **transcode(): void** – Converts media into standardized, web-optimized formats.
- **generateThumbnail(): void** – Produces a thumbnail for image/video previews.
- **attachTranscript(transcriptId: string): void** – Links a transcript to the media item.

Class: Transcript

Attributes:

- **transcriptId**: Unique identifier for the transcript.
- **mediaId**: Foreign key linking to the corresponding media item.
- **language**: Language of the transcript.
- **text**: Full transcript text.
- **timestamps**: Time-aligned text segments for audio/video.
- **generatedBy**: Specifies whether the transcript was human-created or auto-generated.

- **createdAt**: Timestamp for transcript creation.

Methods:

- **search(query: string): List<string>** – Searches for a query inside the transcript.
- **alignTimestamps(): void** – Adjusts the timing alignment with the source media.
- **export(format: string): file** – Exports the transcript in text or subtitle formats.

Class: Scene3D

Attributes:

- **sceneId**: Unique identifier for the 3D scene.
- **communityId**: Foreign key referencing the associated community.
- **sourceMediaList**: Collection of media items used for reconstruction.
- **modelUri**: Storage location of the generated 3D model.
- **methodUsed**: Reconstruction method (3DGS as primary, NeRF as baseline).
- **status**: Current build status (**pending**, **processing**, **ready**, **failed**).
- **metadata**: Camera pose summaries and build parameters.
- **performanceStats**: Metrics such as model size or rendering speed.
- **createdAt**: Timestamp when the reconstruction entry was created.

Methods:

- **startBuild(sourceMediaList: List<string>): void** – Initiates the 3D reconstruction pipeline.
- **getPreview(): uri/file** – Returns a preview of the reconstructed scene.
- **optimizeForWeb(): void** – Compresses and encodes the model for browser rendering.

Class: AuditLog

Attributes:

- **logId**: Unique identifier for the log entry.
- **actorId**: Identifier of the user who performed the action.
- **actionType**: Description of the action (upload, edit, delete, etc.).
- **targetId**: Identifier of the affected resource.

- **timestamp:** Time when the action occurred.
- **details:** Additional metadata such as old/new values or IP address.

Methods:

- **record(actorId: string, actionType: string, targetId: string, details: dict): void** – Creates a new audit log entry.
- **query(filters: dict): List<AuditLog>** – Retrieves log entries that match specific criteria.

4.2 Data Pipeline

This section describes how data flows from user upload to discovery and 3D reconstruction.

4.2.1 Overview (stages)

1. **Ingestion (Client):** Admin uploads media via web form (file + metadata + consent docs). Basic client-side validation runs (file types, size).
2. **API Accept / Validation:** Backend validates metadata schema, consent presence, and user authorization.
3. **Storage:** Raw media stored in Object Store (S3). Backend persists metadata record in DB with ‘status=processing’ for media.
4. **Asynchronous Workers:**
 - **Transcoding/Thumbnailing:** create web-friendly derivatives (images: webp/jpeg smaller; audio: mp3/ogg; video: H.264 mp4).
 - **Transcript Generation:** send audio/video to Speech-to-Text worker (Whisper-like or cloud ASR); produce transcript and timestamps.
 - **Indexing:** push metadata + transcript to Search Index (ElasticSearch) for full-text search.
 - **3D Build (optional):** enqueue scene build job (3D Gaussian Splatting) if user requests a 3D reconstruction from a set of images.
5. **3D Reconstruction:** GPU worker pulls images from storage, computes camera poses (COLMAP or similar), and performs 3D Gaussian Splatting (3DGS) optimization. The system generates a set of Gaussian primitives representing the scene, along with preview renders. Optimized scene data, metadata, and performance statistics are stored in the database.
6. **Delivery / Viewing:** Frontend fetches optimized assets and renders 3D scene (Three.js viewer) and media via CDN.
7. **Audit & Monitoring:** All actions logged to AuditLogs; status updated throughout.

4.2.2 Resilience and observability

- Workers report progress and errors to the API and AuditLogs.
- Retry semantics for transient failures (exponential backoff).
- Storage versioning to preserve original files and derived artifacts.

4.3 Technical Details and Algorithms

4.3.1 3D Gaussian Splatting (3DGS) Reconstruction Pipeline

3D Gaussian Splatting Algorithm

3D Gaussian Splatting (3DGS) is a recent real-time neural rendering technique that represents a scene as a collection of anisotropic Gaussian primitives instead of encoding it in a neural network. This explicit representation enables significantly faster rendering and training compared to Neural Radiance Fields (NeRF), making it suitable for interactive applications.

Scene Representation

In 3DGS, a scene is modeled as a set of Gaussian primitives:

$$G_i = (\mu_i, \Sigma_i, c_i, \alpha_i)$$

where:

- μ_i : 3D position of the Gaussian
- Σ_i : covariance matrix defining shape and orientation
- c_i : RGB color
- α_i : opacity

Each Gaussian contributes to the final rendered image through alpha blending.

Input Requirements

The system requires:

- Multi-view RGB images (20–200 images with overlap)
- Camera intrinsics and extrinsics (computed via COLMAP)

Initialization

A sparse point cloud generated from Structure-from-Motion (SfM) is used to initialize Gaussian positions. Each point is converted into an initial Gaussian primitive.

Optimization

Gaussian parameters are optimized using gradient-based methods to minimize reconstruction error:

$$\mathcal{L} = \sum ||I_{\text{rendered}} - I_{\text{groundtruth}}||^2$$

Parameters updated during optimization:

- Position (μ)
- Covariance (Σ)
- Color (c)
- Opacity (α)

Rendering

Unlike NeRF, which relies on volumetric ray marching, 3DGS uses rasterization-based rendering. Each Gaussian is projected onto the image plane and contributes to pixel color using alpha compositing.

This enables:

- Real-time rendering (30–60 FPS)
- Efficient GPU utilization
- Low latency interaction

3DGS Reconstruction Pipeline

Purpose: reconstruct immersive 3D scenes from multi-angle field photos for real-time virtual exploration.

Inputs:

- Image set for scene (20–200 overlapping images),
- Camera intrinsics and extrinsics (estimated via COLMAP),
- Optional metadata for scene organization.

Outputs:

- Optimized Gaussian-based 3D representation,
- Real-time renderable scene for web viewer,
- Preview images and metadata (camera pose summary, performance stats),
- Status updates and logs.

Steps:

1. Image acquisition and preprocessing
2. Camera pose estimation (Structure-from-Motion)
3. Gaussian initialization from sparse point cloud
4. Optimization of Gaussian parameters
5. Real-time rendering integration (Three.js viewer)

5. Methodology, Experiments and Results OR Testing and Analysis

5.1 Methodology

This project follows an iterative, modular development methodology combining full-stack web development with experimental 3D reconstruction workflows. The system was developed in two parallel components: (1) a digital cultural archive platform and (2) a 3D scene reconstruction module. The workflow integrates data collection, preprocessing, model generation, and deployment into a unified pipeline.

5.1.1 System Development Approach

The development process followed an incremental model, where core archival functionality was implemented first, followed by integration of advanced visualization features. The major phases included:

1. Requirements gathering and domain understanding
2. System architecture and database design
3. Implementation of media archival platform
4. Integration of transcription and search features
5. Experimentation with 3D reconstruction techniques
6. Transition to 3D Gaussian Splatting for real-time rendering
7. Deployment and performance optimization

5.1.2 Data Collection

Cultural data was collected in the form of:

- Photographs of community environments
- Audio interviews and oral histories

- Video recordings of cultural practices
- Contextual metadata and consent documentation

For 3D reconstruction, multi-view images were captured with overlapping viewpoints to ensure sufficient coverage of the scene. Typically, between 20 and 200 images were used for each reconstruction.

5.1.3 Preprocessing Pipeline

Before reconstruction, the collected images underwent preprocessing:

- Image resizing and format normalization
- Removal of blurry or redundant frames
- EXIF extraction for camera information (when available)
- Structure-from-Motion (SfM) using COLMAP for camera pose estimation

The output of this stage included camera intrinsics, extrinsics, and a sparse point cloud used for initialization.

5.1.4 3D Reconstruction Method

Initially, Neural Radiance Fields (NeRF) were explored for scene reconstruction. While NeRF produced high-quality outputs, its slow training time and rendering latency made it unsuitable for real-time web-based deployment. Therefore, the methodology transitioned to 3D Gaussian Splatting (3DGS) as the primary reconstruction approach.

In 3DGS, the scene is represented as a set of Gaussian primitives. The reconstruction process includes:

1. Initializing Gaussians from sparse point cloud
2. Optimizing Gaussian parameters (position, color, covariance, opacity)
3. Minimizing photometric reconstruction error
4. Generating real-time renderable scene representation

This method significantly reduced training time while enabling interactive visualization.

5.1.5 Web Integration

The reconstructed scenes were integrated into the web platform using a browser-based 3D viewer. The integration involved:

- Exporting optimized Gaussian representation
- Converting assets to web-compatible format
- Loading scene via Three.js-based renderer
- Linking scenes to community pages in the archive

Users can navigate reconstructed environments interactively, allowing immersive exploration of cultural spaces.

5.1.6 Search and Metadata Indexing

All uploaded media and transcripts were indexed using a search engine. The indexing pipeline included:

- Metadata extraction
- Transcript generation using speech-to-text
- Full-text indexing
- Tag-based filtering

This enabled efficient retrieval of cultural artifacts and contextual information.

5.1.7 Evaluation Criteria

The system was evaluated based on:

- Reconstruction quality
- Rendering performance
- Scene load time
- Search accuracy
- Usability of archival interface

3D Gaussian Splatting demonstrated improved performance compared to NeRF, particularly in rendering speed and interactivity, making it suitable for deployment in a web-based cultural archive.

6. Conclusion and Future Work

6.1 Conclusion

EthnoVerse presents a comprehensive approach to digital cultural preservation by integrating a scalable archival platform with immersive 3D visualization technologies. The system successfully combines full-stack web development, structured metadata management, and modern 3D reconstruction techniques to create an interactive and accessible repository for marginalized cultural knowledge.

A key contribution of this project is the integration of 3D scene reconstruction into a cultural archive. While Neural Radiance Fields (NeRF) were initially explored for their high-fidelity rendering capabilities, the transition to 3D Gaussian Splatting (3DGS) proved to be a critical design decision. 3DGS enabled significantly faster training and real-time rendering, making it suitable for web-based deployment and interactive user experiences. This shift highlights the importance of aligning technical choices with practical constraints such as performance, scalability, and accessibility.

The system demonstrates that cultural archives need not remain static repositories; instead, they can evolve into dynamic, experiential platforms that allow users to engage with cultural environments in meaningful ways. By linking media, transcripts, and reconstructed spaces, EthnoVerse provides a richer, multi-modal representation of community knowledge.

Beyond its technical contributions, the project emphasizes ethical considerations in digital archiving, including informed consent, contextual metadata, and responsible representation. These aspects are essential when working with historically underrepresented communities.

Overall, EthnoVerse establishes a strong foundation for future work at the intersection of computer science, digital humanities, and cultural preservation.

6.2 Future Work

While the current system achieves its core objectives, several avenues exist for further enhancement and expansion.

6.2.1 Scalability and Multi-Community Support

Future iterations of the system can extend beyond a single community to support multiple cultural groups. This would require improved data organization, scalable storage solutions, and more robust indexing mechanisms to handle larger datasets.

6.2.2 Mobile-Based Data Collection

Developing a mobile application for data collection would enable field researchers and community members to capture and upload media directly. This would streamline the data ingestion process and encourage participatory archiving.

6.2.3 Enhanced 3D Reconstruction

Although 3D Gaussian Splatting enables real-time rendering, further improvements can be explored:

- Hybrid approaches combining 3DGS with mesh extraction for better geometry
- Improved handling of dynamic scenes and moving subjects
- Optimization for low-resource devices such as mobile browsers

6.2.4 User Interaction and Immersive Features

Future work can enhance user engagement by introducing:

- Guided virtual tours within reconstructed environments
- Annotation layers for cultural storytelling
- Virtual or augmented reality (VR/AR) support

6.2.5 AI-Assisted Metadata and Search

The system can be extended using advanced AI techniques:

- Automatic tagging of images and videos using computer vision
- Semantic search using embeddings instead of keyword matching
- Multilingual transcription and translation support

6.2.6 Community Participation and Ethics

A key direction for future work is enabling community-driven contributions:

- User-submitted content with moderation workflows
- Consent management dashboards
- Tools for community members to curate and narrate their own archives

These improvements would not only enhance the technical robustness of EthnoVerse but also strengthen its role as a participatory and ethically grounded cultural preservation platform.

7. Reflection

7.1 Reflections

7.1.1 Individual Reflections

This project provided a deeply interdisciplinary learning experience, combining technical development with critical engagement in cultural representation and archival practices. On the technical side, we gained hands-on experience in full-stack development, including database design, API development, and frontend integration. Additionally, working with modern 3D reconstruction techniques introduced us to complex concepts in computer vision and neural rendering, which required significant self-learning beyond standard coursework.

One of the most challenging aspects was understanding and implementing 3D reconstruction pipelines. Initially, Neural Radiance Fields (NeRF) were explored, which required grappling with volumetric rendering, ray sampling, and neural network optimization. While conceptually rich, NeRF proved difficult to adapt for real-time applications. Transitioning to 3D Gaussian Splatting (3DGS) required not only learning a new technique but also rethinking the system design to prioritize performance and usability.

Beyond technical challenges, this project also strengthened our ability to work with real-world, imperfect data. Issues such as inconsistent image quality, incomplete metadata, and ethical considerations around consent required careful handling. This experience highlighted the importance of designing systems that are robust, flexible, and socially responsible.

Overall, the project helped us develop both technical depth and critical awareness, particularly in understanding how computational tools intersect with cultural knowledge and representation.

7.1.2 Team Reflection

The project was carried out collaboratively, requiring clear communication, coordination, and division of responsibilities. Tasks were distributed based on individual strengths, with team members focusing on areas such as backend development, frontend design, and 3D reconstruction.

Regular meetings and progress updates were essential in maintaining alignment across different components of the system. However, coordinating work across parallel modules—particularly the archival platform and the 3D reconstruction pipeline—posed challenges. Integration required continuous communication and iterative adjustments.

One key learning from the team experience was the importance of flexibility. As the project evolved, especially during the transition from NeRF to 3DGS, roles and responsibilities had to be adapted. This required openness to feedback and a willingness to rework previously completed components.

Despite these challenges, the collaborative process was productive and allowed us to build a more comprehensive system than would have been possible individually. The experience reinforced the value of interdisciplinary teamwork, especially in projects that span both technical and conceptual domains.

7.1.3 Process Reflection

The iterative development approach proved to be effective for this project. Building the archival platform first provided a stable foundation upon which more experimental features, such as 3D reconstruction, could be integrated.

One aspect that worked particularly well was the modular system design. By separating components such as media processing, transcription, search indexing, and 3D reconstruction, we were able to develop and test each module independently before integration.

However, one area that could be improved is early-stage technical validation. Initially, more time was spent exploring NeRF without fully accounting for its computational limitations in a web-based setting. Earlier benchmarking and feasibility analysis could have reduced the need for later redesign.

Another challenge was managing the complexity of integrating diverse technologies, including cloud storage, GPU-based processing, and frontend rendering. Debugging across these layers required significant time and coordination.

Overall, the process highlighted the importance of balancing experimentation with practicality, especially when working with emerging technologies.

7.1.4 Plans vs Achievements

At the proposal stage, the project aimed to develop a digital cultural archive enhanced with immersive 3D scene reconstruction using Neural Radiance Fields (NeRF). The goal was to create high-quality, photorealistic representations of community environments.

In practice, while the archival platform was successfully implemented as planned, the 3D reconstruction component underwent a significant shift. NeRF, although effective in producing high-quality results, was found to be computationally expensive and unsuitable for real-time interaction. As a result, the project transitioned to 3D Gaussian Splatting (3DGS), which enabled faster training and real-time rendering.

This change led to several outcomes:

- **Achieved:**
 - Fully functional archival platform with media management and search
 - Integration of transcription and metadata indexing
 - Real-time 3D scene visualization using 3DGS

- **Modified:**
 - Replacement of NeRF with 3DGS as the primary reconstruction method
- **Added:**
 - Performance-focused optimization for web-based rendering
 - Improved integration of 3D scenes into the user interface

The decision to shift from NeRF to 3DGS was driven by practical constraints such as rendering speed, hardware limitations, and the need for interactive user experiences. This reflects an important lesson from the project: design decisions must balance theoretical potential with real-world feasibility.

While not all initial plans were implemented exactly as proposed, the final system is more aligned with the project’s core objective of accessibility and usability. The challenges encountered ultimately led to a more robust and deployable solution.

Appendix A. Code

Our code can be found at <https://github.com/habib-university/Kaavish-Template>.

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